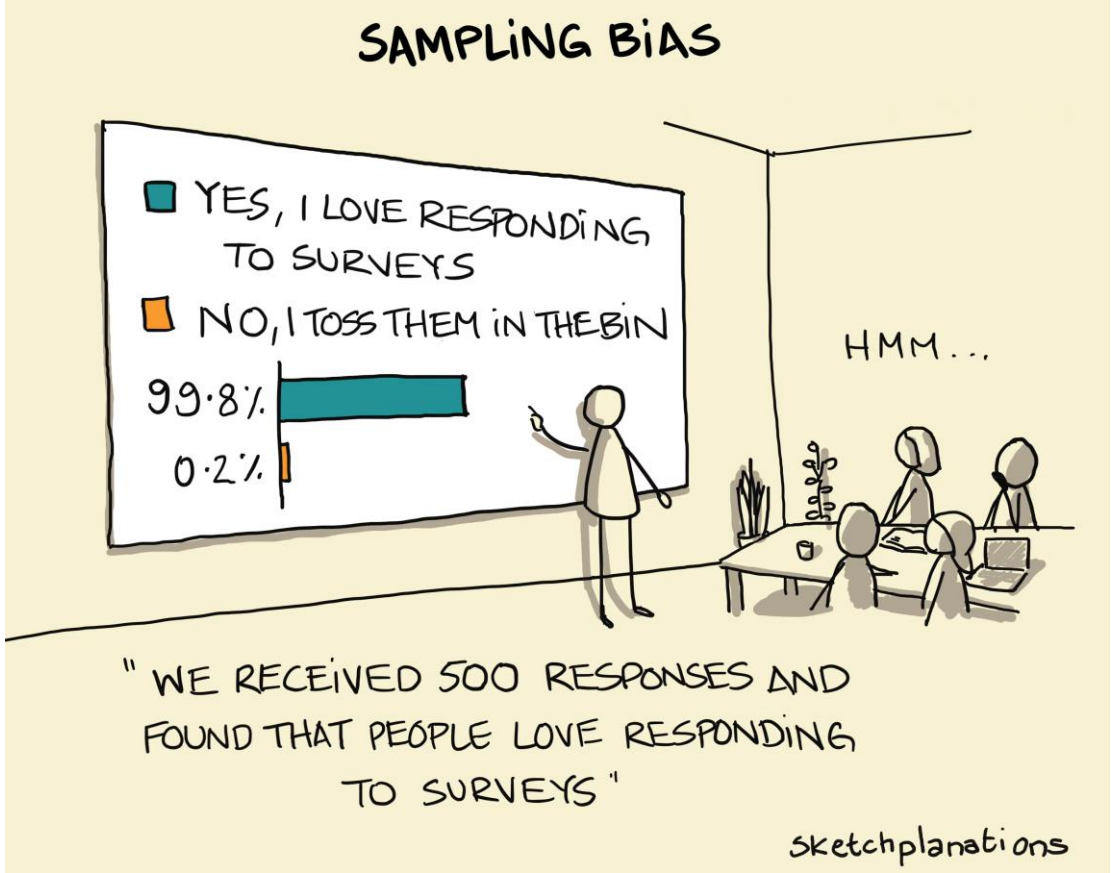
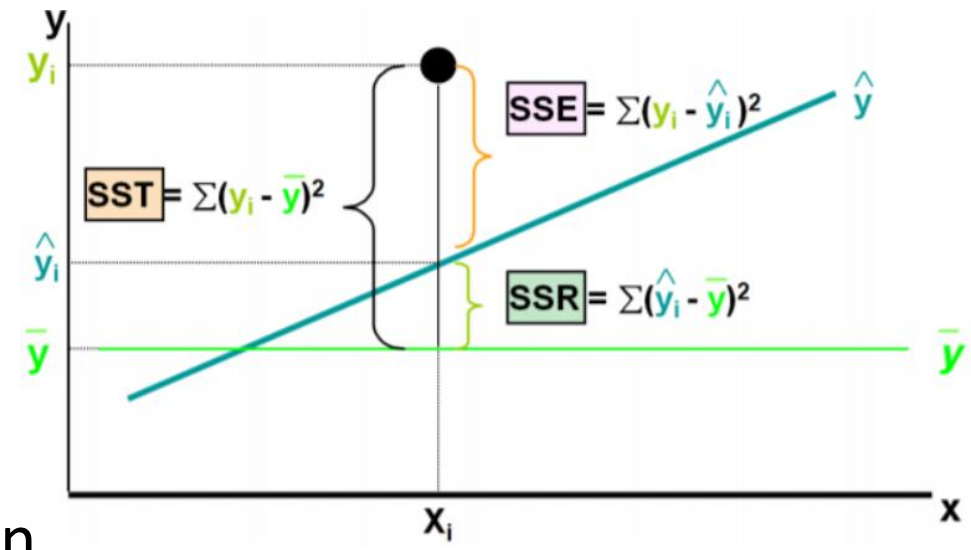
Best of the Rest

Chapter 15, 16 and beyond



Sums of Squares in Regression

- *Total Sum of Squares (SST)*: the total variation in the observed values of the response variable
  - $SST = \sum(y_i - \bar{y})^2$
- *Regression Sum of Squares (SSR)*: the variation in the observed values of the response variable explained by the regression:
  - $SSR = \sum(\hat{y}_i - \bar{y})^2$
- *Error Sum of Squares (SSE)*: the variation in the observed values of the response variable not explained by the regression:
  - $SSE = \sum(y_i - \hat{y}_i)^2$



Standard Error of Estimate

- A measure of the scatter, or dispersion, of the observed values around the line of regression. S_e
- Describes how precise the prediction of Y is based on X, or how inaccurate the estimate might be.

$$S_e = \sqrt{\frac{SSE}{n-2}}$$

Interpretation of SSE Value

Small

- Data is relatively close to the regression line
- Regression equation can be used to predict Y with little error

Large

- **Data are widely scattered around the regression line**
- **Regression equation will not provide a precise estimate Y**

Inferences for Linear Regression

- Assumptions underlying linear regression
  - The standard deviations of these normal distributions are the same
  - The Y values are statistically independent. This means that in the selection of a sample, the Y values chosen for a particular X value do not depend on the Y values for any other X values
- We can compute Confidence intervals and Prediction intervals for regression analyses:
  - Confidence Interval: reports the mean value of Y for a given X

$$\mathbf{Y' \pm t(s_{y.x})\sqrt{\frac{1}{n} + \frac{(X - \overline{X})^2}{\sum (X - \overline{X})^2}}}$$

  - Prediction Interval: reports the values of Y for a particular value of X

$$\mathbf{Y' \pm t(s_{y.x})\sqrt{1 + \frac{1}{n} + \frac{(X - \overline{X})^2}{\sum (X - \overline{X})^2}}}$$

Inferences for Linear Regression

- Hypothesis tests for the slope of the population regression line

Suppose that the variables *x* and *y* satisfy the assumptions for regression inferences. Then, for samples of size *n*, each with the same values $x_1, x_2, \ldots, x_n$ for the predictor (independent) variable, the variable

$$t = \frac{b_1 - \beta_1}{s_e / \sqrt{s_{xx}}}$$

Has the t-distribution with df = n-2

The Regression t-Test

- Purpose: perform a hypothesis test to decide whether a predictor variable is useful for making predictions.
- Assumptions: the underlying normal regression assumptions hold
- Perform the 5 step process with

Step 1: The null and alternative hypotheses are, respectively,

$$H_0\colon\ \beta_1 = 0$$
$$H_1\colon\ \beta_1 \neq 0$$

Step 3: The test statistic is:

$$t = \frac{b_1}{s_e \big/ \sqrt{S_{xx}}}$$

With df = n-2

Example

Consider the data of the number of hours of light per day, X and the length of the breeding season (days) for photoperiodism in waterfowl. The data are also included in the file *breed.csv* on D2L.

Perform regression analysis in Minitab, and determine using $\alpha = 0.05$ whether the hours of light in a day is a true predictor of the length of the breeding period.

Hours, X	Breed, Y
12.8	110
13.9	54
14.1	98
14.7	50
15.0	67
15.1	58
16.0	52
16.5	50
16.6	43
17.2	15
17.9	28

Example Solutions

Inferences in Correlation

- Purpose: to perform a hypothesis test for a population linear correlation coefficient, ρ.
- Assumptions: the underlying normal regression assumptions hold
- Perform the 5-step process with

Step 1: The null and alternative hypotheses are, respectively,

$$H_0\colon\ \rho = 0$$
$$one\ of\ H_1\colon \rho \neq 0, \rho < 0, \rho > 0$$

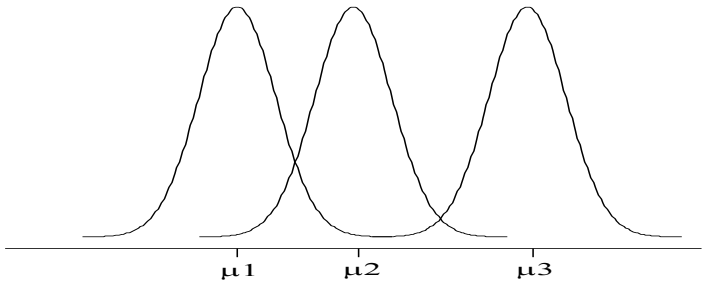
Step 3: The test statistic is:

$$t = \frac{r}{\sqrt{\frac{1 - r^2}{n - 2}}}$$

With df = n-2

ANOVA (Analysis of Variance)

- Performs a test of multiple comparisons of population means
- Assumptions: we have *k* independent simple random samples from normal distributions with equal variances/standard deviations.
- We need to take into account the variation within samples (selecting different samples from within the same population); and the variation between samples (across different populations)



ANOVA

- The *treatment mean square (MSTR)* is the variation among the sample means $MSTR = SSTR/(k-1)$ where SSTR is the treatment sum of squares and k is the number of populations under consideration.
- The *error mean square (MSE)* is the variation within the samples: $MSE = \frac{SSE}{n-k}$, where SSE is the error sum of squares and n is the total number of observations.
- *F-statistic (F)* is the ratio of the variation among the sample means to the variation within the samples: $F = MSTR/MSE$

ANOVA Test

- Purpose: to perform a hypothesis test to compare *k* population means
- Assumptions:
  - Simple random samples
  - Independent samples
  - Normal populations
  - Equal population standard deviations
- Step 1: The null and alternative hypotheses are, respectively

$$H_0\colon \mu_1 = \mu_2 = \mu_3 = \cdots = \mu_k$$
$$H_a\colon not\ all\ means\ are\ equal$$

- Step 3: compute the value of the test statistic $F = \dfrac{MSTR}{MSE}$ by completing the table:

ANOVA TABLE

Source of Variation	***df***	**Sum of Squares *SS***	**Mean Square** ***MS = SS/df***	**Test Statistic** ***F = MSTr/MSE***
Treatments	k - 1	*SSTr*	*MSTr*	F_o =
Error	N - k	*SSE*	*MSE*	
Total	N - 1	*SSTo*		

Example

A school of education conducted a study to compare four methods of teaching reading comprehension (method A is the standard method). Twenty-four grade 5 students of similar ability were randomly assigned to the four methods. After four months of instruction, a standardized reading test (SRT) was given to the students. Their scores are given in the file *oneway.csv* on D2L. Is there evidence at the 0.05 significance level to conclude that the mean SRT scores differ for students taught by different teaching methods?

Example Solutions